

# CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

## End-Semester Examination

Roll Number: \_\_\_\_\_

Name: \_\_\_\_\_

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.  
All the best!

## Formulae for Reference

1. [75 points]

Consider  $N$  images  $\{x^n\}_{n=1}^N$  of semantic segmentations obtained from MRI scans, of the chest, from  $N$  different individuals. The associated MRI scans are unavailable. Each semantic segmentation contains voxels with  $L = 4$  different labels corresponding to the (i) lungs (say, label 1), (ii) heart (say, label 2), (iii) bone (say, label 3), and (iv) all other voxels labeled as background (say, label 0).

**Assumption 1:** Assume all images employ a **consistent** scheme for labeling objects in images, e.g., the object labeled 1 in all images is indeed the lungs.

**Assumption 2:** Assume all images lie within a **common** anatomical coordinate frame.

**Assumption 3:** Assume that there exists a **single** representative probabilistic segmentation image  $\bar{x}$  for all the  $N$  individuals such that it indicates the probability of any voxel  $v$ , in the common anatomical coordinate frame, belonging to a class with label  $l$ .

(a) (12 points)

Under Assumption 1, Assumption 2, and Assumption 3, answer the following.

- { 8 points: 4 + 4 } Mathematically define a quantitative measure of dissimilarity  $d(\cdot, \cdot)$  between (i) the representative image  $\bar{x}$  and (ii) a semantic segmentation image  $x^n$ .

Use  $d(\cdot, \cdot)$  to mathematically formulate an optimization problem to find  $\bar{x}$ , where the solution can be obtained in closed form.



- { 4 points } Derive the closed-form solution to solve the aforementioned optimization problem to find the optimal  $\bar{x}$ .

(b) (15 points)

For the same problem as in Question 1a, now relax Assumption 1 and assume that there isn't guaranteed consistency in the labeling scheme across semantic-segmentation images  $x^n$ , e.g., image  $x^1$  may have label 1 for the lungs, but image  $x^2$  may have label 3 for the lungs. Assume that you don't have the expertise or the budgets needed to correct the labels manually.

- { 2 points } Describe, in words, how the optimal  $\bar{x}$  would be undesirably affected when the  $N$  images aren't using a consistent labeling scheme.

- { 5 points } Mathematically define a quantitative measure of dissimilarity  $D(\cdot, \cdot)$  between (i) the representative image  $\bar{x}$  and (ii) a semantic segmentation image  $x^n$ , such that the measure isn't affected by the labeling scheme used for  $x^n$  and that the measure can be computed exactly. Assume that any manual or rule-based relabeling is impossible.

- { 3 points } Use  $D(\cdot, \cdot)$  to mathematically formulate an optimization problem to find  $\bar{x}$ .

- { 5 points } Provide an implementable algorithm to solve the aforementioned optimization problem to find the optimal  $\bar{x}$ .

(c) (18 points)

For the same problem as in Question 1a, relax Assumption 1, as done in Question 1b. In addition, now relax Assumption 2 and assume that the  $N$  images aren't in the same anatomical coordinate space.

- { 2 points } Describe, in words, how the optimal  $\bar{x}$  would be undesirably affected when the  $N$  images aren't in the same anatomical coordinate space.

- { 4 points } To overcome the undersirable effects, mathematically re-design a new dissimilarity measure  $\Delta(\cdot, \cdot)$  and reformuate the optimization problem for finding the  $\bar{x}$ .

- { 12 points } Provide an implementable algorithm to solve the aforementioned optimization problem to find the optimal  $\bar{x}$ .





(d) (30 points)

For the same problem as in Question 1a, relax Assumption 1, as done in Question 1b, and relax Assumption 2, as done in Question 1c. In addition, now relax Assumption 3 and assume the  $N$  images actually belong to  $K > 2$  clusters (e.g., based on disorders) and that, instead of finding a single representative  $\bar{x}$ , you want to cluster the images into  $K$  clusters and estimate  $K$  cluster representatives  $\bar{x}^k$  using the EM algorithm. Within EM, assume that, for each  $x^n$ , the a-priori probability of belonging to any cluster is  $1/K$ .

- { 4 points } Propose a strategy to model the probability density function (PDF) for each cluster based on the dissimilarity function  $\Delta(\cdot, \cdot)$  you designed in Question 1c.

- { 15 points } Given the PDF model, specify an implementable EM algorithm for the clustering.

Specify the entities that EM considers as “hidden variables”.

Specify the entities that EM considers as “parameters”.

Specify the  $Q(\cdot; \cdot)$  function considered by EM.

Within each iteration of EM, specify optimization algorithms for all variables involved.

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- { 11 points } Now your goal is to modify the EM algorithm you just described to incorporate a spatial-smoothness prior on all cluster representatives  $\bar{x}^k$ .  
Specify the entities that EM considers as “hidden variables”.  
Specify the entities that EM considers as “parameters”.  
Specify the prior model.  
Within each EM iteration, specify the modified optimization problem(s) to be solved involving the  $Q(\cdot; \cdot)$  function; an optimization algorithm isn’t required.

2. [15 points]

Suppose you want to align two pointsets, each representing shape, each with  $N$  points lying in a 2-dimensional physical domain. Suppose you decide to keep one pointset  $\{x_n \in \mathbb{R}^2\}_{n=1}^N$  fixed (as given as input), and then solve for all parameters underlying a similarity transformation to align the other pointset  $\{y_n \in \mathbb{R}^2\}_{n=1}^N$  to the fixed pointset.

- (5 points) Mathematically formulate an optimization problem for the task.

- (10 points) Specify an algorithm for solving the optimization problem, and mathematically derive the analytical formulae for closed-form estimates for all parameters underlying the similarity transformation.

3. [10 points]

- (3 points) What is the physical quantity that MRI (which produces a stack of 2D images) measures (as we covered in class) ?

- (3 points) What are the two main quantities that the Bloch equations relate/connect ?

- (4 points) Specify the mathematical model, underlying MRI, between the measured signals and the physical quantity being measured.





